



# Providing open forest structural characteristics for high conservation priority wildlife species in southeastern U.S. pine plantations

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## ABSTRACT

The decline of early successional and open forests and their wildlife inhabitants has resulted in increased efforts to understand and conserve these communities. Natural and anthropogenic disturbances that historically created and maintained open forest conditions have been disrupted, necessitating wildlife use of alternative habitat sources (e.g., pine plantations, utility rights-of way, abandoned agricultural fields) to carry out their life history. We reviewed available literature to estimate the range of structural conditions suitable for four open forest species in regional decline and compared these ranges to available structure in loblolly pine (*Pinus taeda*) forests managed for economic return. Species included Bachman's sparrow (*Aimophila aestivalis*), northern bobwhite (*Colinus virginianus*), prairie warbler (*Dendroica discolor*), and eastern diamondback rattlesnake (*Crotalus adamanteus*). We estimated habitat availability and connectivity for each of these four species during a 60-year simulation on an economically- and operationally-feasible pine landscape (> 22,000 ha) managed for sawtimber production in the West Gulf Coastal Plain. Habitat requirements, including minimum patch size and dispersal constraints, of Bachman's sparrow, northern bobwhite, prairie warbler, and eastern diamondback rattlesnake are generally met by pine stands in stand establishment and thinned, mid-rotation stages. However, habitat availability in pine plantations may be highly ephemeral, tends to occupy the upper end of basal area and canopy closure tolerance, and may be more suitable for open woodland species (e.g., Bachman's sparrow) than grassland species (e.g., northern bobwhite). The range of habitat associations in this literature review highlights the need to refine targets of structural metrics identified by open pine restoration initiatives to encompass the full range of conditions occupied by open forest species. Current literature is strongly biased to avian habitat associations. Few papers explore habitat associations of herpetofauna of conservation concern in the southeastern U.S., and these species tend to be less mobile and therefore, more vulnerable to landscape changes and the ephemeral nature of open vegetative structure in pine plantations.

## 1. Introduction

Early successional forests, savannas, and woodlands (collectively known as open forest (Hanberry and Thompson, 2019, Hanberry et al., 2014)) are characterized by open canopies, a sparse to moderate midstory layer, and a dense herbaceous (i.e., graminoids, forbs, ferns) understory, often maintained by frequent disturbances. Open forests have historically included shrubland, glades, beaver meadows, abandoned agricultural fields, and regenerating stands following catastrophic disturbance (e.g., hurricane, forest fire; Askins, 2001). Disruption of historic natural and anthropogenic disturbance patterns via fire suppression, beaver removal, and mesophication has led to widespread decline in open forests across the eastern United States (Nowacki

and Abrams, 2008, Fowler and Konopik, 2007, DeStefano and Deblinger, 2005, Van Lear and Harlow, 2002, Trani et al., 2001). The fire-dependent longleaf pine (*Pinus palustris*) ecosystems have declined from 60 million acres in pre-settlement southeastern States to 2.95 million acres by 1995 (Outcalt and Sheffield, 1996). Early successional forests have declined to 5.5% of their original extent since the mid-twentieth century, with evident declines in 22 of 31 eastern U.S. states (King and Schlossberg, 2014). Regional decline of early successional forests in the southeastern U.S. averages 3% per year (King and Schlossberg, 2014), but local declines and proportions of area in early successional condition varies widely across the region (e.g., early successional forest occupies < 6% of forested area in Kentucky and nearly 40% in Mississippi; Trani et al., 2001). Concurrent early successional,

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shrubland, and pine-grassland wildlife population declines have increased the conservation priority of these communities (Schlossberg et al., 2010, Hunter et al., 2001). For example, 43% of shrubland birds were identified as high-concern by the Canadian Wildlife Service (Dunn et al., 1999), and several of the top-ranked birds of conservation concern (“Red Watch List”) by Partners in Flight are associated with early successional and savanna conditions (e.g., Gunnison sage grouse [*Centrocercus minimus*], black-capped vireo [*Vireo atricapillus*], Florida scrub-jay [*Aphelocoma coerulescens*], Bachman’s sparrow [*Aimophila aestivalis*]; Rosenberg et al., 2016). Forest management promoting open forest wildlife species is necessary to mitigate habitat loss and alteration. Alternative, man-made early successional habitat sources (e.g., utility rights-of-way (Askins et al., 2012, King et al., 2009b, King and Byers, 2002), wildlife and silvicultural openings (King et al. 2009a), reclaimed surface mines (Bulluck and Buehler, 2006)) may be needed, but not relied upon, to meet breeding and wintering habitat needs (Schlossberg and King, 2015, King and Schlossberg, 2014, Schlossberg et al., 2010).

Modern silviculture can play an important role in the provision of open forests for resident and migratory species. Silvicultural activities such as timber removal, prescribed fire, and selective herbicide create structural characteristics such as appropriate vegetation height, a herbaceous understory dominated by graminoids and forbs, and an open midstory layer, which are defining aspects of open forests (Iglay et al., 2014, King and Schlossberg, 2014). Prairie warblers (*Dendroica discolor*), indigo buntings (*Passerina cyanea*), and northern bobwhite (*Colinus virginianus*) have been observed using recently harvested stands following replanting (Iglay et al., 2018, Owens et al., 2014, Morris et al., 2013, Hanberry et al., 2012, Iglay et al., 2012, Lane et al., 2011a, Jones et al., 2009b, LeGrand et al., 2007). Open forest bird species also inhabit thinned, mid-rotation stands for brief (1–4 years) periods (Iglay et al., 2018, Singleton et al., 2013, White et al., 1999, King et al., 1998, Tucker et al., 1998). Prescribed fire with or without herbicide can further extend stages of open vegetative structure by reducing or removing midstory hardwoods and shrubs (Singleton et al., 2013, Sladek et al., 2008, White et al., 1999, King et al., 1998, Tucker et al., 1998).

We examined the potential for managed pine landscapes to provide open forest conditions. To address how economically-managed pine forests in the southeastern U.S. can provide open structural conditions for complementing regional priority conservation concerns (e.g., open forest wildlife), we reviewed literature and quantitatively summarized structural characteristics in regenerating (1–5 years after planting) and mid-rotation (thinning to final timber harvest) loblolly pine (*P. taeda*) stands. We then reviewed literature to determine habitat associations of four species identified as high conservation priorities at the national and state levels that are associated with open forests: Bachman’s sparrow, northern bobwhite, prairie warbler, and eastern diamondback rattlesnake (*Crotalus adamanteus*; Rosenberg et al., 2016, Georgia Department of Natural Resources, 2015, Holcomb et al., 2015, The National Bobwhite Technical Committee, 2011). Finally, we estimated amount and connectivity of suitable habitat in a planted pine landscape simulated to meet operational and economic objectives for sawtimber management over 60 years. Although gopher tortoise (*Gopherus polyphemus*) is native to open pine ecosystems and does occur within planted pine landscapes, we did not include gopher tortoise in our analysis due to a similar effort recently published (see Greene et al., 2019).

## 2. Methods

We gathered information from peer-reviewed literature regarding structural characteristics of loblolly pine stands and habitat associations for species of conservation concern to determine habitat availability in a simulated forest landscape. We conducted a systematic literature search to identify structural and vegetation conditions (e.g., canopy closure, vegetation height, graminoid coverage) of even-aged loblolly

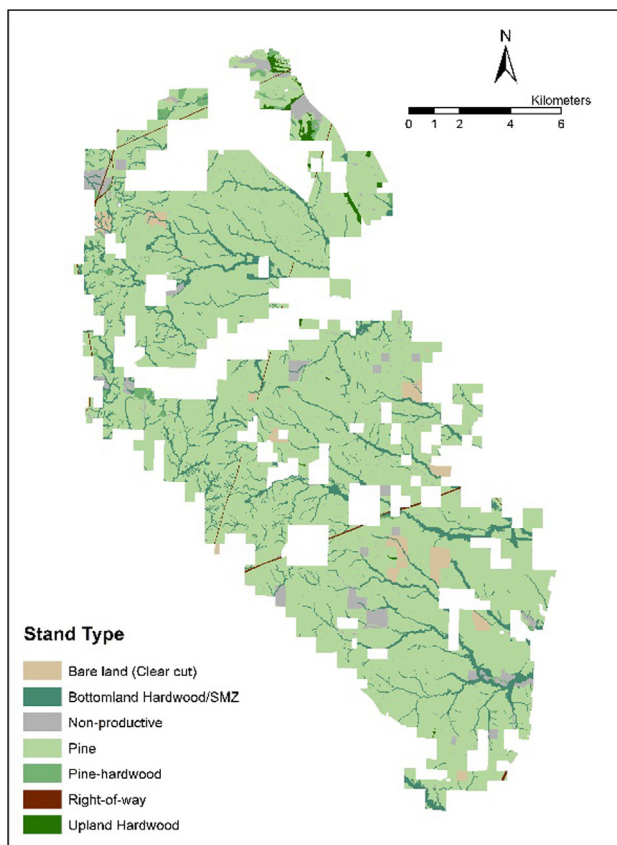
pine stands managed for sawtimber production during stand establishment and mid-rotation phases. We identified the stand establishment period as immediately following timber removal to approximately 5 years old when pine seedlings grow into canopy closure and understory vegetation transitions to a shade-tolerant community (Lane et al., 2011b, Jones et al., 2009a). During stand establishment, common management practices (e.g., herbaceous weed control, fertilizing) support seedling survival and productivity. We identified mid-rotation as the stage from thinning (pre-commercial or commercial thinning) until final harvest. Despite inclusion of what might be considered late-rotation in our description, we wanted to include the entire time when mid-rotation management practices, such as thinning, selective herbicide application, and prescribed fire, may be applied to create open forest conditions (Iglay et al., 2014).

We searched academic databases and information service providers (e.g., EBSCOhost, Google Scholar, JSTOR, ProQuest Dissertations and Theses, Science Direct, SpringerLink, USDA Forest Service Treesearch, Wildlife and Ecology Studies Worldwide, Wiley Online Library) using open forest, stand establishment, and mid-rotation search terms (e.g., early successional, early seral, regeneration, open pine, *Pinus taeda*, thinning). We supplemented our search with knowledge of relevant literature. We restricted our literature search to the southeastern U.S. Coastal Plains due to consistency of management practices in this ecoregion that may differ from other areas in the Southeast (e.g., Piedmont ecoregion) and interest in regional wildlife species of conservation concern. We synthesized available data and calculated summary statistics for 11 structural and vegetative metrics during stand establishment: pine diameter at breast height (DBH) and percent coverage of overstory canopy, midstory shrubs, herbaceous understory, grasses, ferns, forbs, vines, woody plants, debris, and bare ground (Gulf Coastal Plains and Ozarks Landscape Conservation Cooperative, 2013). A summary of structural metrics of mid-rotation loblolly pine stands was published separately (see Greene et al., 2019).

Because inclusion of numerous stand establishment practices could increase bias and inconsistencies in summary statistics, we restricted data to mechanical or chemical site preparation and banded application of herbicide applied one year after loblolly pine seedlings were planted. Due to the rapid growing conditions of regenerating pine stands, we limited our inclusion of stand establishment studies to the first 5 years after initial site preparation, and we expected variation in structural metrics based on stand age, land use history, soils, and other factors. Studies reporting structural conditions in pine stands did not have equal sample sizes with *n* ranging from 1 to 45 study sites. Therefore, we used weighted means to quantitatively summarize structural conditions. Studies with higher sample sizes received relatively more weight in the averaging process.

We then searched for publications about our four focal species of conservation concern in the Southeastern Coastal Plains [Bachman’s sparrow, northern bobwhite, prairie warbler, and eastern diamondback rattlesnake] to determine vegetation characteristics associated with each species. We chose these species based on conservation status, data availability, and documented occurrence in managed pine stands. We originally planned to include a broader array of herpetofauna (e.g., gopher frog [*Rana capito*], Louisiana pine snake [*Pituophis melanoleucus ruthveni*], pygmy rattlesnake [*Sistrurus miliarius*]), but too few empirical habitat structure associations precluded these additional species. We included publications across a range of cover types, ownerships, and management regimes. We searched the same 11 databases used for our structural metric literature review but replaced search terms with species’ common and scientific names. We documented associations between forest management and the resulting vegetation structure, with species occurrence, abundance, and nesting success. We also tracked number of studies contributing to each vegetation metric and relationship to species, and summarized data deficiencies and knowledge gaps for species when evident.

We next obtained a GIS coverage of an existing managed pine



**Fig. 1.** Initial stand composition in a managed pine landscape (22,270 ha) located in the West Gulf Coastal Plain used to simulate 60 years of business-as-usual forest management for sawtimber pine production.

landscape located in the West Gulf Coastal Plain from the National Council for Air and Stream Improvement, Inc. to estimate the amount and connectivity of suitable habitat in a planted pine landscape. All georeferences, soil data, and existing planting densities were removed prior to the release of landscape data to protect proprietary information. The initial landscape composition consisted of 22,270 ha of which 84% (18,722 ha) was in managed pine, 2% (427 ha) was recently clearcut and not yet replanted with pine, and 9% (1963 ha) was in bottomland hardwood streamside management zones (SMZs; Fig. 1). Managed pine stands ( $n = 1216$ ) ranged in age from 0 to 30 years. Initial age class distribution of managed pine stands approximated a normal distribution ( $\bar{x} = 13.6$  years,  $\sigma = 7.2$  years). Site indices of planted pine stands averaged 18.1 m at base age 25 ( $SD = 1.9$  m). The landscape had an additional 208 ha in managed pine-hardwood stands ranging from 6 to 66 years, and 118 ha in upland hardwood stands ranging from 3 to 89 years ( $< 1\%$  of landscape).  $> 3\%$  of the landscape was deemed unproductive for timber, and 85 ha were in right-of-way easements.

We simulated forest management using Habplan, a spatially-explicit harvest scheduling program (Van Deusen, 2001, 1999). Habplan incorporates user-defined objective functions for spatially-explicit problems and provides allowable management regimes for each polygon (hereafter, stand). Habplan uses a simulation-based approach using the Metropolis Algorithm to identify optimized or near optimal solutions that meet objective function criteria (Van Deusen, 2001, 1999). After a 30-year training period, we simulated forest management for 60 years (approximately two timber rotations) to determine economic and habitat suitability solutions. We averaged results from the top 5 solutions for all estimations of economic feasibility and habitat suitability. We set commercial thinning when trees were 13–18 years and final harvest via clearcut at 25–32 years, following recommendations from the National

Council for Air and Stream Improvement, Inc. regarding standard sawtimber management practices. We did not thin pine-hardwood or upland hardwood stands, but harvested them when trees were 40–50 years old. We replanted all stands to loblolly pine at 1075 stems/ha one year following harvest in bedded rows, a common planting density and site preparation practice in the southeastern U.S. (personal communication, J. M. Guldin, U.S.D.A. Forest Service, Southern Research Station; D. A. Miller, Weyerhaeuser Company; T. B. Wigley, National Council for Air and Stream Improvement, Inc.). Our landscape complied with voluntary sustainable forestry certification which stipulates that (1) average clearcut size cannot exceed 50 ha, except in special circumstances (e.g., regulatory, ecological, or forest health objectives), and (2) a stand cannot be clearcut if an adjacent stand had been clearcut within 3 years and dominant tree height has not reached 1.5 m (Sustainable Forestry Initiative, 2015). Although the Sustainable Forestry Initiative (2015) supports wildlife conservation, primary principles and objectives focus on forest stewardship, not best management practices for promoting biological diversity. To ensure economic feasibility of management regimes, we constrained harvest scheduling by a volume evenness goal ( $\leq 10\%$  deviation allowed between consecutive years) and maximized total volume cut for 40% of pine stands (i.e., final harvest scheduled 32 years after planting).

We used the first 30 years of simulation to train the landscape to our management and spatial constraints. We estimated habitat availability at 5-year intervals during years 31–90 of the simulation (i.e., 60-year simulation). Habitat availability was categorized into three classes: open forest conditions at stand establishment and mid-rotation, or other (e.g., closed canopy pine stands, SMZs). Utility rights-of-way (ROWs) comprised  $< 1\%$  ( $\sim 85$  ha) of our landscape area but could disproportionately influence connectivity of open conditions by functioning as dispersal corridors (Amaral et al., 2016, Jones et al., 2008, Bolger et al., 2001). Therefore, we calculated habitat availability with and without the inclusion of ROWs to address how corridors within managed landscapes affect spatial connectivity of larger open patches.

We calculated class and landscape spatial metrics in Fragstats 4.4 (McGarigal et al., 2012) to quantify the connectedness of each condition class (stand establishment, mid-rotation, and other) for each 5-year interval and reported means and standard deviations for each spatial metric across the 60-year simulation. We calculated total area, percent of landscape, number of patches, and mean patch size for each condition class to assess simple composition of our landscape. We used four metrics of connectivity for each class: (1) Euclidean distance (straight-line distance between patches in the same condition class), (2) aggregation, (3) clumpiness indices which measure dispersion (extent to which patches are clumped), and (4) interspersed-juxtaposition index which measures interspersed (extent to which patch types are intermixed). We calculated the contagion index, which subsumes dispersion and interspersed, for the entire landscape to assess aggregation without reference to any particular condition class. All connectivity indices herein are reported as percentages with values near 0 indicating disaggregation and/or low interspersed and values near 100 indicating high degrees of spatial clumping and/or intermixing.

### 3. Results & discussion

#### 3.1. Stand establishment metrics in pine plantations

We found 20 publications providing structural and vegetative metrics of loblolly pine stands in the first 5 years of stand establishment from 1987 to 2015. Study sites were located throughout the southeastern U.S.: Arkansas ( $n = 1$ ), Georgia ( $n = 1$ ), Louisiana ( $n = 4$ ), Mississippi ( $n = 5$ ), North Carolina ( $n = 4$ ), South Carolina ( $n = 1$ ), and Texas ( $n = 5$ ). The most common structural and vegetative metrics were percent coverage of forbs, woody plants, grasses, vines, and bare ground and total herbaceous understory coverage. In contrast, few data points were available to estimate overstory pine DBH and percent shrub

**Table 1**

Summary statistics for structural and vegetative metrics during the first 5 years of stand establishment in planted loblolly pine forests in the Southeastern Coastal Plains. Canopy closure was estimated using spherical densiometers. Understory percent coverages were determined using line-intercept and belt-transect sampling techniques, or in a radius around a randomly-generated point. Groundcover is > 100% if growth-forms overlapped.

| Metric                    | Stand establishment summary statistics <sup>a</sup> |            |   |
|---------------------------|-----------------------------------------------------|------------|---|
|                           | Weighted Mean $\pm$ SD                              | Range      | n |
| Pine DBH (cm)             | 6.6 $\pm$ 2.3                                       | 03.9–18.9  | 2 |
| Canopy Cover (%)          | 48.8 $\pm$ 26.8                                     | 11.0–066.3 | 5 |
| Shrub (%)                 | 19.3 $\pm$ 14.6                                     | 01.5–056.0 | 3 |
| Herbaceous Understory (%) | 66.3 $\pm$ 30.6                                     | 17.2–107.7 | 7 |
| Grass (%)                 | 32.1 $\pm$ 17.5                                     | 15.2–092.6 | 7 |
| Fern (%)                  | 3.1 $\pm$ 3.3                                       | 00.0–011.7 | 5 |
| Forb (%)                  | 29.3 $\pm$ 17.0                                     | 05.6–082.2 | 9 |
| Vine (%)                  | 14.1 $\pm$ 20.1                                     | 01.1–072.2 | 6 |
| Woody (%)                 | 19.4 $\pm$ 16.2                                     | 03.0–071.0 | 9 |
| Debris (%)                | 11.8 $\pm$ 12.0                                     | 01.9–058.7 | 5 |
| Bare Ground (%)           | 34.1 $\pm$ 16.7                                     | 02.9–053.3 | 6 |

<sup>a</sup> Sources: Fritts et al. (2015), Hanberry et al. (2013), Posner (2012), Grace (2011), Lane et al. (2011b), Novak (2011), Jones et al. (2010), Jones et al. (2009a), Bechard (2008), LeGrand et al. (2007), Edwards et al. (2006), Humphries (2005), Thill and Koerth (2005), Mihalco (2004), Miller et al. (2004), Duda (2003), Strong (1999), Mitchell et al. (1995), Dickson et al. (1993), Burnett (1987).

coverage.

Following site preparation and planting, the non-pine vegetative community was dominated by shade-intolerant, pioneer species such as blackberries (*Rubus* spp.) and graminoids before transitioning to a sparser, shade-tolerant understory as canopy coverage increases. This successional trajectory was evident in summary statistics from stand establishment studies (Table 1). Total herbaceous understory coverage averaged 66%, and grasses and forbs had the greatest percent coverage of all growth forms. Canopy coverage averaged 49%, and many studies observed that canopy closure generally exceeded 50% once trees

reached 5 years of age (e.g., Lane et al., 2011b, LeGrand et al., 2007). All metrics of open forest condition during stand establishment varied substantially. This was likely because of the dynamic successional phases observed during stand establishment with range extremes represented by years 1 and 5. Other possible sources inflating variation were edaphic factors, land use history, long-term weather patterns (e.g., drought), and changes in management practices from 1987 to 2015, among other factors.

### 3.2. Mid-rotation stand metrics in pine plantations

In a review of 42 publications of structural conditions in managed loblolly pine stands at mid- and late-rotation stages, we found that pine stands approximate open pine structure in “good” condition for up to four years after a commercial thinning (Greene et al., 2019). Additional management actions (e.g., prescribed fire, selective herbicide) can extend open pine structure in “good” to “fair” condition for an additional four years. We quantitatively summarized basal area, diameter at breast height (DBH), canopy cover, midstory hardwood cover and basal areas, and coverages of midstory shrubs, total herbaceous understory, graminoids, forbs, vines, and understory woody vegetation in a previous publication (Greene et al., 2019), and found that thinned, mid-rotation stands average a pine basal area of 17 m<sup>2</sup>/ha, canopy closure of 84%, hardwood basal area of 6 m<sup>2</sup>/ha, midstory shrub cover of 21%, and understory herbaceous cover of 35%. In the first three years following a commercial thin, pine stands averaged understory vegetative coverages of 17% graminoids, 26% vines, 12% forbs-ferns-legumes, and 9% woody plants. Additional mid-rotation management increased graminoid, forbs-ferns-legumes, and woody growth forms, but decreased vine coverage.

### 3.3. Habitat associations of focal species

We found 50 publications providing data about habitat characteristics for four focal species inhabiting open conditions (e.g., open pine woodlands and grasslands, pine-oak savannas, open-structured pine

**Table 2**

Habitat characteristics of four open forest wildlife species of regional conservation concern in the Southeastern Coastal Plains.

|                                      | Bachman's sparrow <sup>a</sup>     | Northern bobwhite <sup>b</sup>    | Prairie warbler <sup>c</sup>                                                                | Eastern diamondback rattlesnake <sup>d</sup> |
|--------------------------------------|------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------|----------------------------------------------|
| Basal area (m <sup>2</sup> /ha)      | < 20                               | < 20                              | < 8.5                                                                                       | 1–10                                         |
| Pine DBH (cm)                        | n.d. <sup>e</sup>                  | n.d.                              | Small-diameter trees; probably selecting lack of overstory canopy, not a specific DBH class | Sparse large trees                           |
| Canopy closure                       | 40–60%, but can tolerate up to 80% | < 60%, but can tolerate up to 80% | < 60%, but can tolerate up to 75%                                                           | 10–60%                                       |
| Midstory shrubs                      | < 20%, but may tolerate up to 50%  | 10–40%                            | 10–50%                                                                                      | < 30%                                        |
| Herbaceous understory                | > 40–50%                           | > 20%                             | 25–50%                                                                                      | 20–60%                                       |
| Grass <sup>f</sup>                   | > 40–50%                           | n.d.                              | 25–50%                                                                                      | n.d.                                         |
| Bare ground                          | < 35%                              | ~10%                              | n.d.                                                                                        | n.d.                                         |
| Minimum patch size (ha) <sup>g</sup> | 5                                  | 20                                | 1–3                                                                                         | 60                                           |
| Distance to nearest patch (km)       | < 3                                | 1                                 | Inconclusive data                                                                           | < 5                                          |

<sup>a</sup> Bachman's sparrow studies: Fish et al. (2019), McIntyre et al. (2019), Fish et al. (2018), Fish (2017), Hannah et al. (2017), Jones et al. (2017), Pickens et al. (2017), Taillie et al. (2015), Allen and Burt (2014), Kronenberger et al. (2014), Barrioz et al. (2013), Jones et al. (2013), Brooks and Stouffer (2010), Stober and Kremetz (2006), Tucker et al. (2004), Conner et al. (2002), Haggerty (1998), Plentovich et al. (1998), Tucker et al. (1998), Wilson et al. (1995), Brooks (1987), McKittrick (1979).

<sup>b</sup> Northern bobwhite studies: McIntyre et al. (2019), Butler et al. (2017), Hannah et al. (2017), DeMaso et al. (2014), Crosby et al. (2013), Singleton et al. (2013), Marsh et al. (2012), Lane et al. (2011a,b), Iglay (2010), Burke et al. (2008), Riddle et al. (2008), Rader et al. (2007), Lusk et al. (2006), Greenfield et al. (2003), Cram et al. (2002), Michener et al. (2000), Taylor and Burger (2000), Wilson et al. (1995).

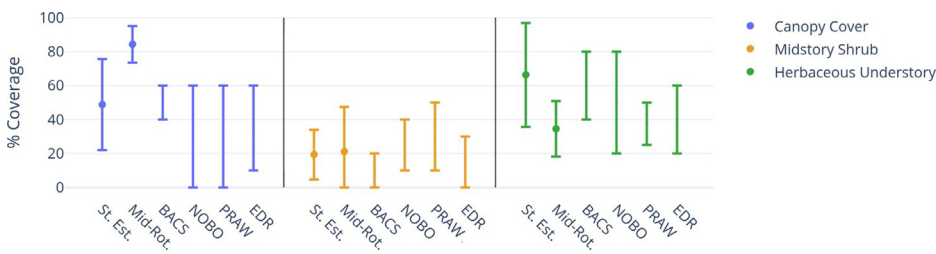
<sup>c</sup> Prairie warbler studies: Reidy and Thompson (2018), Jones et al. (2017), Roberts and King (2017), Reidy et al. (2014), Barrioz et al. (2013), Akresh (2012), Shake et al. (2011), Comer et al. (2011), Wilson et al. (1995), Tonkovich and Stauffer (1994), Brooks (1987).

<sup>d</sup> Eastern diamondback rattlesnake studies: McIntyre et al. (2019), Fill et al. (2015b), Surdick (2013), Waldron et al. (2006), Kain (2003), Timmerman (1995).

<sup>e</sup> No data.

<sup>f</sup> Grass is a subset of herbaceous understory.

<sup>g</sup> Based on reported home range sizes and density estimates.



**Fig. 2.** Canopy, midstory shrub, and herbaceous understory coverages provided in stand establishment (SE) and thinned, mid-rotation (MR) pine stands are compared to habitat characteristics of four focal species: Bachman's sparrow (BACS), northern bobwhite (NOBO), prairie warbler (PRAW), and eastern diamondback rattlesnake (EDR). Midpoints indicate mean condition in stand establishment and mid-rotation stands. For focal species, midpoints indicate the midpoint of the range of suitable conditions (Table 2). Error bars indicate either standard deviation (SE and MR) or range of suitable conditions (focal species).

plantations) with a focus on sites in the Southeastern Coastal Plains. Additionally, we reviewed studies that provided context to habitat associations but lacked data concerning specific thresholds or ranges of suitable structure (i.e., basal area, DBH, canopy closure, midstory shrubs, understory composition; e.g., Fill et al., 2015a, Bried et al., 2011, Dunning and Watts, 1990). Several studies documented responses of focal species to forest management (e.g., Morris et al., 2013, Tucker et al., 2004, Burger et al., 1998), woodland and savanna restoration (e.g., Roach et al., 2019), agricultural programs (e.g., Evans et al., 2013, Potter et al., 2011), and urbanization (e.g., Schlossberg et al., 2011, Gifford et al., 2010).

Using the 50 publications providing data about habitat components, we offer possible ranges and thresholds of structural metrics and patch size for each focal species (Table 2). We found that focal species associations with structural metrics were often within the range provided by regenerating and/or commercially thinned, mid-rotation pine stands (Fig. 2). Focal species often had strong habitat associations linked to frequent disturbance regimes (e.g., prescribed fire, hurricanes; Pickens et al., 2017, Akresh et al., 2015, Barrios et al., 2013, Brooks and Stouffer, 2010, Duda, 2003). The greatest canopy coverage observed during stand establishment (66%) was within the range inhabited by Bachman's sparrow, northern bobwhite, and prairie warbler, but several studies demonstrated that lower canopy coverages (e.g., < 60%) have stronger associations with occupancy (Pickens et al., 2017, Taillie et al., 2015, Allen and Burt, 2014, Crosby et al., 2013, Akresh, 2012, Tucker et al., 1998) and likely reflect a preference for open overstory structure. In regenerating pine stands, midstory coverage of trees < 3 years old may be sufficient to sustain focal species that prefer a shrub component (e.g., northern bobwhite and prairie warbler) without deterring species who avoid a midstory presence (e.g., Bachman's sparrow). While total herbaceous understory coverage was moderate to high in regenerating pine stands, a lack of consistency in coverage of specific growth forms (e.g., graminoids) and bare ground made evaluation of regenerating and thinned pine stands difficult in some cases (e.g., assessing high grass cover needs of Bachman's sparrow).

Some seemingly significant relationships between focal species and structural components in Table 2 may reflect species' selection of a vegetative community or disturbance regime rather than selection of each individual habitat component. For example, associations of Bachman's sparrow with bare ground may not reflect selected use of these habitat components; rather, certain ranges of bare ground co-occur with the basal area, canopy closure, and midstory shrub components that Bachman's sparrow seem to be selecting.

Bachman's sparrow was most often correlated with canopy and understory grass coverages. Bachman's sparrow exhibited a quadratic relationship with canopy cover with occupancy greatest when canopy cover ranged from 40% to 60%, but Bachman's sparrow was also present in stands of greater canopy coverage (Hannah et al., 2017, Pickens et al., 2017, Allen and Burt, 2014). Bachman's sparrow occurrence decreased with increasing shrub-layer density and height (Allen and Burt, 2014), and optimal midstory shrub coverage is likely to be less than 20% (McIntyre et al., 2019, Taillie et al., 2015). Although Bachman's sparrow has inhabited stands undergoing pine-grassland

restoration with greater shrub coverage (e.g., 49%; Plentovich et al., 1998), the use of such stands may be indicative of marginal habitat in the surrounding landscape. A dense herbaceous understory dominated by graminoids and forbs appeared to be a key habitat requirement for Bachman's sparrows. Graminoid and forb coverage was positively associated with abundance (Allen and Burt, 2014, Conner et al., 2002, Allen, 2001, Plentovich et al., 1998) and song rate (Allen, 2004). However, positive correlations with the herbaceous understory may reach a saturation point in environments with very high graminoid and forb coverages wherein further increases in coverages no longer generate increases in abundance (Taillie et al., 2015, Brooks and Stouffer, 2010). For example, Haggerty (1995) found no differences between nesting and non-nesting sites when grass coverage averaged 66% and 73%, respectively. Distance to nearest suitable habitat (< 3 km) was also an important consideration often included in Bachman's sparrow occupancy models (e.g., Pickens et al., 2017, Taillie et al., 2015). Although canopy closure in thinned, mid-rotation pine stands is greater than in other land cover types inhabited by Bachman's sparrow, mid-rotation stands still met the midstory shrub and herbaceous understory components of Bachman's sparrow habitat (Fig. 2). Thus, mid-rotation stands may be briefly suitable immediately after thinning when maximum light reaches the forest floor and stimulates growth of herbaceous cover.

Northern bobwhite abundance was strongly influenced by canopy cover (< 60%), midstory shrub cover (10–40%), vertical stratification of groundcover, and bare ground cover (~10%; McIntyre et al., 2019, Cram et al., 2002, Taylor and Burger, 2000). In the Southeastern Coastal Plains, occupancy and foraging success were associated with low midstory shrub cover (10–25%; McIntyre et al., 2019, Burke et al., 2008, Taylor and Burger, 2000). In the Great Plains and Rio Grande Plains ecoregions, sites become more suitable for bobwhite nesting as shrub cover increases (e.g., in excess of 25%; DeMaso et al., 2014, Lusk et al., 2006), and may reflect the importance of shrub cover to daytime loafing, escape cover, and winter cover. For vertical stratification, Davis (1964) observed that northern bobwhite roosting sites were located in erect herbaceous vegetation, usually bunchgrasses, at ~1 m height. More recent studies have reiterated the importance of dense clumps of herbaceous vegetation near the ground for northern bobwhites (Arredondo et al., 2007, Rader et al., 2007, Conner et al., 2002, Townsend et al., 2001).

Ground layer metrics received less attention than herbaceous understory, but debris and bare ground coverage were important to understanding horizontal patchiness in understory vegetation. Similarly, bare ground coverage influenced nest-site selection by northern bobwhites with sites becoming more suitable when exposed ground was less than 30% (Lusk et al., 2006). Rader et al. (2007) found that bobwhite nest sites averaged 11% bare ground compared to 25% at paired random locations. However, northern bobwhites can inhabit stands with greater bare ground exposure (e.g., 75%) if other appropriate vegetative structure is available (Tonkovich and Stauffer, 1994). Habitat structure enabling successful chick foraging of arthropods is consistent with aforementioned abundance relationships with herbaceous, woody, and bare ground coverage. Proximity of bare ground to

**Table 3**

Class and landscape metrics (mean  $\pm$  SD) for stand establishment (i.e., 1–5 years after planting), mid-rotation (i.e., 1–4 years after thinning), and other conditions (e.g., closed canopy pine stands, SMZs) in loblolly pine stands located in the West Gulf Coastal Plain. Metrics were calculated during 60 years of simulated forest management. Stand establishment metrics were calculated with and without utility rights-of-way (ROWs).

|                                                    | Stand Establishment with ROWs | Stand Establishment without ROWs | Mid-rotation     | Other            |
|----------------------------------------------------|-------------------------------|----------------------------------|------------------|------------------|
| Total area (ha)                                    | 13,995<br>(2544)              | 13,625<br>(2545)                 | 13,620<br>(3005) | 69,432<br>(3727) |
| % of landscape                                     | 14.4<br>(2.6)                 | 14.0<br>(2.6)                    | 14.0<br>(3.1)    | 71.5<br>(3.8)    |
| Number of patches                                  | 177.8<br>(22.2)               | 123<br>(15.0)                    | 122.4<br>(15.5)  | 155.9<br>(30.5)  |
| Mean patch size (ha)                               | 79.3<br>(14.5)                | 111.0<br>(17.5)                  | 112<br>(24)      | 446<br>(118)     |
| Euclidean distance (m)                             | 357<br>(29.7)                 | 447<br>(47.0)                    | 451.6<br>(60.8)  | 194.2<br>(3.8)   |
| Aggregation Index (%) <sup>a</sup>                 | 79.7<br>(1.4)                 | 81.4<br>(2.8)                    | 80.6<br>(1.7)    | 91.4<br>(0.8)    |
| Clumpiness Index (%) <sup>a, b</sup>               | 76.2<br>(1.2)                 | 77.6<br>(1.1)                    | 77.5<br>(1.5)    | 69.6<br>(1.5)    |
| Interspersion-Juxtaposition Index (%) <sup>c</sup> | 47.6<br>(6.0)                 | 45.0<br>(5.8)                    | 50.1<br>(6.2)    | 98.6<br>(1.1)    |

<sup>a</sup> Index ranges from 0% (maximally disaggregated) to 100% (maximally clumped).

<sup>b</sup> Percentage for clumpiness index traditionally given in decimal form; multiplied by 100 for this table.

<sup>c</sup> Index ranges from 0% (a patch type is adjacent to only one other patch type) to 100% (a patch type is maximally interspersed and juxtaposed with all other patch types).

denser nesting cover is critically important to northern bobwhite, and may be of especial importance to successful chick foraging post-hatch (Burke et al., 2008, Taylor and Burger, 2000). Finally, northern bobwhite occupancy is highly influenced by proximity to other populations, and managers might consider a maximum distance of 1 km between habitat patches, depending on permeability of the surrounding matrix (Crosby et al., 2013).

Studies measuring response of populations and food plants to management practices provided additional support for northern bobwhite habitat structure relationships. Mid-rotation pine stands receiving applications of selective herbicide and prescribed burning have increased abundance and density of northern bobwhites compared to burn-only, herbicide-only, and control plots (Butler et al., 2017, Singleton et al., 2013, Iglay, 2010). Mechanical disturbance (e.g., disking, stand establishment during pine plantings), prescribed burning, and selective herbicide are appropriate management actions to improve understory plant diversity and increase coverage of northern bobwhite food plants (Marsh et al., 2012, Masters, 2007, Greenfield et al., 2003, Greenfield et al., 2002).

Probability of prairie warbler occurrence and abundance had positive relationships with seedlings and saplings (< 8 cm), but this association is likely a by-product of prairie warblers selecting sites with open overstory canopy and a midstory shrub layer rather than specific DBH classes (Fink et al., 2006, Aquilani, 2002, Allen, 2001, Dickson and Segelquist, 1979). Large-diameter trees may also be suitable if an open canopy and midstory shrub component is retained. Prairie warbler may tolerate higher canopy cover (up to 75%) than suspected of an open canopy-associated bird. However, the upper threshold is probably in the range of 60–75% (Comer et al., 2011). Prairie warblers also appeared to be associated with lower (< 50%) and absent canopy coverage more often than canopy coverage of 50–75% (Akresh et al., 2015, Perry and Thill, 2013, Brooks, 1987). Prairie warbler had the strongest associations with midstory shrubs of any focal species with a likely

optimal range of 10–50% coverage (Reidy and Thompson, 2018, Roach et al., 2018). Prairie warblers also briefly inhabit regenerating clearcuts in which young trees form a pseudo-shrubby layer that often exceeds 50% (Hanberry et al., 2012, Lane, 2011a, 2011b, Edwards et al., 2006, Brooks, 1987).

Few studies assessed habitat associations of eastern diamondback rattlesnake despite the need to understand occupancy patterns and reproductive success outside remnant longleaf pine (*P. palustris*) tracts (e.g., Baxley and Qualls, 2009). Like other focal species, eastern diamondback rattlesnake was associated with savannas (Fill et al., 2015b, Waldron et al., 2008, Steen et al., 2007). Waldron et al. (2008) found that the probability of use by eastern diamondback rattlesnake increased 1.5 times for every 25% decrease in canopy cover at a pine savanna preservation site maintained with prescribed fire. Habitat suitability modeling using long-term vegetation and faunal data in southwestern Georgia suggested that eastern diamondback rattlesnake occupy stands with lower basal area, canopy closure, and shrub and herbaceous understory coverages than observed for our other focal species (McIntyre et al., 2019). Radio telemetry data and occupancy modelling have demonstrated that while eastern diamondback rattlesnakes are primarily associated with pine-grassland communities, they can use other land cover types (e.g., mixed pine hardwood, mesic and xeric hammock; Steen et al., 2012, Lee, 2009, Timmerman, 1995) and seem to respond positively to a limited hardwood component (Steen et al., 2012, Waldron et al., 2008). Males typically had larger home ranges (84.8 ha, 59.5 ha, and 158.9 ha) than non-gravid females (28.6 ha, 8.2 ha, and 88.8 ha; Waldron et al., 2006, Kain, 2003, and Timmerman, 1995, respectively) with density estimates of 1 individual per 5–8 ha (Timmerman, 1995, Means, 1986).

### 3.4. Landscape analysis

Approximately 28% of the landscape we studied may provide habitat for open forest species based on analysis of open conditions in economically-managed loblolly pine stands in the West Gulf Coastal Plain. Availability of open forest was evenly distributed between stand establishment and thinned, mid-rotation pine stands (Table 3). Mean open forest patch size averaged 80 ha and 112 ha in stand establishment and thinned, mid-rotation pine, respectively, during 60 years of forest management simulation. Patch size of available open forest condition exceeds minimum patch requirements for sustaining populations of our focal species. Euclidean distances between open forest patches in stand establishment were less than the straight-line distances needed to travel between thinned, mid-rotation stands. While distances in both stand types were less than the distance to nearest suitable patch documented in our focal species literature review, species may take a circuitous route and thus, travel much further than the Euclidean distance. Aggregation and clumpiness index values indicated that open stands were highly aggregated. Aggregation of open stands was unexpected in a landscape managed according to Sustainable Forestry Initiative standards which prohibit clearcutting if an adjacent stand has pine trees less than three years of age and seedling height less than 1.5 m. However, our analysis defined stand establishment as the five-year period following planting. Thus, our analysis allowed one-year old stands to be adjacent to four- and five-year old stands while maintaining Sustainable Forestry Initiative certification. Stand establishment and thinned, mid-rotation pine stands were also moderately interspersed and juxtaposed with each other and other stand types. This intermixing of stand establishment and thinned, mid-rotation stands further decreases the effort of focal species dispersing to adjacent available habitat.

## 4. Conclusion

Habitat suitability requirements of Bachman's sparrow, northern bobwhite, prairie warbler, and eastern diamondback rattlesnake are generally met by pine stands in stand establishment and, to a lesser

extent, thinned, mid-rotation stages. However, habitat availability in pine plantations may be highly ephemeral depending on management regime (e.g., planting density, herbicide use, thinning intensity) and site index (Greene et al., 2019, Iglay et al., 2012, Lane et al., 2011a,b). Rapid structural changes in growing pine stands and thinned, mid-rotation stands demonstrates that managed pine landscapes can present a shifting mosaic of short-term (1–5 years) suitable habitat (Greene et al., 2019) without achieving the long-term, “ideal” habitat suitability one might find in pine-grassland communities managed with frequent fire or open forest managed for biodiversity on protected lands.

Although our literature review of structural conditions in pine plantations generally align with desired open forest structural conditions, structure of pine plantations is not equally suitable for all open forest species, and specific species-habitat relationships may be met by either stand establishment or mid-rotation stands but not always by both stand types. Thinned, mid-rotation stands tend to occupy the upper end of basal area and canopy closure targeted for maintaining open forest species (McIntyre et al., 2019, Nordman et al., 2016), a product of less frequent disturbance by tree harvest compared to the frequent, low-intensity fires that historically maintained grasslands, shrublands, and open forest in the southeastern U.S. (Van Lear et al., 2005). Thus, thinned, mid-rotation stands may be more suitable for those species that could alternatively be classed as woodland species (e.g., Bachman’s sparrow) while stand establishment conditions may better meet requirements of wildlife classified as early successional forest, pine-grassland, and shrubland species (e.g., eastern diamondback rattlesnake, prairie warbler). Moreover, pine-grassland and woodland species may utilize edges between mid-rotation and regenerating pine stands to compensate for the greater canopy closure in the interior of mid-rotation stands (e.g., gopher tortoises [Lamb et al., 2013, Diemer Berish and Moore, 1993]).

Minimum patch size and distance to nearest suitable patch were met in our pine landscape following SFI adjacency restrictions for our avian focal species. However, this result is not surprising given that our focal avian species are not particularly dispersal-limited (Jones et al., 2017, Taillie et al., 2015). Although our pine management simulation and landscape analysis demonstrated how pine plantations can provide conditions reminiscent of open forests and suitable for open forest species, there is uncertainty about how consistently pine plantations provide habitat for open forest species across the southeastern region given the variability in management regimes, edaphic factors, and legacy influences of prior land use. Caution should be exercised when extrapolating results of our pine landscape simulation to pine plantations outside the Southeastern Coastal Plains.

Current literature is strongly biased to avian habitat associations. Few papers explore habitat associations, patch size, and dispersal ability of herpetofauna of conservation concern, including eastern diamondback rattlesnake, in the Southeastern Coastal Plains. Rather, herpetofauna publications often address diversity and abundance responses to restoration activities (Cantrell et al., 2013, Perry et al., 2009, Ledvina, 2008, Litt et al., 2001) and forest management regimes (MacGowan et al., 2017, Rota et al., 2017, Sanspree, 2016, Sutton et al., 2013). Studies that do report habitat associations often focus on assemblage responses (e.g., species diversity and richness) and landscape structure rather than species-specific habitat characteristics within home ranges (Wolf et al., 2016, Smith, 2011, Barrett and Guyer, 2008, Loehle et al., 2005, Russell et al., 2002, Metts et al., 2001). This lack of information to gauge habitat suitability and composition is problematic as many of these species tend to be less mobile and therefore, more vulnerable to landscape changes and the ephemeral nature of habitat in pine plantations.

Minimum patch size, effect of edge, and efficacy of corridors remain contested issues in the literature. Several papers express concerns about avian use of utility rights-of-way due to area sensitivity, increased edge, and greater brood parasitism that may lead to corridors being population sinks, especially in areas with heavier development (Askins et al.,

2012, Lehnen, 2008, Rodewald and Vitz, 2005, Askins, 1994). However, corridors may benefit shrubland birds by connecting or expanding their habitat availability (King and Byers, 2002, Askins, 1994), and Conservation Reserve Program field borders and food plots have increased usable space for northern bobwhites when implemented in agriculture-dominated landscapes (Evans et al., 2015, Evans et al., 2013, Smith and Burger, 2009, Michener et al., 2000). Although many studies document positive associations between patch size and avian density, species richness, and/or diversity (e.g., Greenberg and Lanham, 2001), minimum patch size recommendations are difficult to establish due to species differences in dispersal ability. In addition, the paucity of data regarding spatial requirements and dispersal-limiting factors, particularly for eastern diamondback rattlesnakes, add to the difficulty of assessing the collective impacts of corridors, patch size, and fragmentation.

Habitat associations of our four focal species in the Southeastern Coastal Plains were similar to habitat associations used in adjacent geographies (e.g., Central Hardwoods Bird Conservation Region, Tirpak et al., 2009). Several initiatives have promoted open pine restoration (e.g., America’s Longleaf, East Gulf Coastal Plain Joint Venture, Gulf Coastal Plains and Ozarks Landscape Conservation Cooperative [GCPO LCC]), and hypothesized desired ecological state accompanied by targets for structural and configuration metrics (e.g., GCPO LCC 2013). While initiatives expect a range of successional conditions and anticipate increases in open forest species as a result of open pine restoration, the structural metric targets for open pine are often skewed toward optimal structural ranges that may not be available in high quantities in landscape matrices of privately-owned land. For example, GCPO LCC (2013) targets canopy cover of < 50% and herbaceous cover of > 65%. However, open forest bird species can successfully reproduce at higher canopy cover and lower herbaceous cover (Loman et al., 2018, Roach et al., 2018, Akresh, 2012, Barber et al., 2001), indicating that targets set by initiatives may miss near-optimal habitat opportunities. In addition, open forest bird species have historically inhabited several vegetative types (e.g., grasslands, shrublands, early successional forests, and open and closed woodlands), but restoration and management targets are often skewed toward savanna-like and early successional forest structure. Thus, there is risk of excluding woodlands at the upper end of tree stocking and canopy cover spectrums (i.e., up to 75% stocking or 80% canopy closure; Dey et al., 2017, Hanberry et al., 2014) that are still suitable for many open forest species. That said, long-term assessment of population trajectories in habitats that skirt the fringe of optimal conditions is warranted to better understand contributions of these habitats to overall populations.

## 5. Limitations

Most studies of habitat associations and structural conditions in pine stands focused on basal area, canopy coverage, and herbaceous understory total and growth-form coverages. Relatively few data points were available for overstory pine DBH, bare ground coverage and distance to nearest suitable patch. Therefore, we acknowledge a degree of uncertainty in the habitat characteristics of focal species presented in Table 2. Additionally, our review of focal species’ habitat characteristics was subjective, as too few data points and differences in studies’ methods precluded a quantitative meta-analysis. While we used the landscape analysis to evaluate how well-managed pine stands meet minimum patch size and dispersal distance limitations of focal species, we were unable to evaluate provision of structural characteristics during the landscape analysis. Thus, we are reliant on stand structure relationships to pine age and management actions in the literature to draw conclusions about the sustained availability of habitat for focal species at the landscape (> 22,000 ha) scale.

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